JAMA | Original Investigation

Effect of Oral Alfacalcidol on Clinical Outcomes in Patients Without Secondary Hyperparathyroidism Receiving Maintenance Hemodialysis
The J-DAVID Randomized Clinical Trial

The J-DAVID Investigators

IMPORTANCE Patients with chronic kidney disease have impaired vitamin D activation and elevated cardiovascular risk. Observational studies in patients treated with hemodialysis showed that the use of active vitamin D sterols was associated with lower risk of all-cause mortality, regardless of parathyroid hormone levels.

OBJECTIVE To determine whether vitamin D receptor activators reduce cardiovascular events and mortality in patients without secondary hyperparathyroidism undergoing hemodialysis.

DESIGN, SETTING, AND PARTICIPANTS Randomized, open-label, blinded end point multicenter study of 1289 patients in 207 dialysis centers in Japan. The study included 976 patients receiving maintenance hemodialysis with serum intact parathyroid hormone levels less than or equal to 180 pg/mL. The first and last participants were enrolled on August 18, 2008, and January 26, 2011, respectively. The final date of follow-up was April 4, 2015.

INTERVENTIONS Treatment with 0.5 μg of oral alfacalcidol per day (intervention group; n = 495) vs treatment without vitamin D receptor activators (control group; n = 481).

MAIN OUTCOMES AND MEASURES The primary outcome was a composite measure of fatal and nonfatal cardiovascular events, including myocardial infarctions, hospitalizations for congestive heart failure, stroke, aortic dissection/rupture, amputation of lower limb due to ischemia, and cardiac sudden death; coronary revascularization; and leg artery revascularization during 48 months of follow-up. The secondary outcome was all-cause death.

RESULTS Among 976 patients who were randomized from 108 dialysis centers, 964 patients were included in the intention-to-treat analysis (median age, 65 years; 386 women [40.0%]), and 944 (97.9%) completed the trial. During follow-up (median, 4.0 years), the primary composite outcome of cardiovascular events occurred in 103 of 488 patients (21.1%) in the intervention group and 85 of 476 patients (17.9%) in the control group (absolute difference, 3.25% [95% CI, −1.75% to 8.24%]; hazard ratio, 1.25 [95% CI, 0.94-1.67]; P = .13). There was no significant difference in the secondary outcome of all-cause mortality between the groups (18.2% vs 16.8%, respectively; hazard ratio, 1.12 [95% CI, 0.83-1.52]; P = .46). Of the 488 participants in the intervention group, 199 (40.8%) experienced serious adverse events that were classified as cardiovascular, 64 (13.1%) experienced adverse events classified as infection, and 22 (4.5%) experienced malignancy-related serious adverse events. Of 476 participants in the control group, 191 (40.1%) experienced cardiovascular-related serious adverse events, 63 (13.2%) experienced infection-related serious adverse events, and 21 (4.4%) experienced malignancy-related adverse events.

CONCLUSIONS AND RELEVANCE Among patients without secondary hyperparathyroidism undergoing maintenance hemodialysis, oral alfacalcidol compared with usual care did not reduce the risk of a composite measure of select cardiovascular events. These findings do not support the use of vitamin D receptor activators for patients such as these.

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Patients with more advanced chronic kidney disease (CKD) are at a higher risk of cardiovascular disease (CVD), particularly patients requiring hemodialysis.[1,2] The additional risk is explained by impairment in traditional and nontraditional risk factors[1] associated with CKD. Mineral and bone disorder (MBD) in CKD[3] is 1 of the nontraditional risk factors.[4] Observational studies[5,6] showed that serum phosphate, calcium, and intact parathyroid hormone (PTH) levels were associated with increased mortality risk in patients undergoing hemodialysis.

Impaired activation of vitamin D by the kidneys may be causatively involved in the pathogenesis of CKD-MBD and increased cardiovascular risk in CKD. A review[7] of cohort studies revealed that the use of active vitamin D sterols, or vitamin D receptor activators (VDRAs), was associated with a lower risk of all-cause mortality,[8] CVD-related mortality,[9] and incident CVD[10] in patients treated with hemodialysis. In a stratified analysis,[8] lower mortality in VDRA users was observed in patients receiving hemodialysis, regardless of serum intact PTH levels. Clinical[11] and experimental studies[12,13] showed potentially beneficial effects of VDRAs, such as suppression of the renin-angiotensin system; modulation of immune functions; anti-inflammatory effects; antiatherosclerotic properties; inhibition of cardiac hypertrophy; and increase of proteins potentially protective against arterial calcification,[14] including fetuin-A and klotho.[15]

However, the potential benefit of VDRA treatment in patients with CKD receiving hemodialysis has not been examined in randomized trials. Thus, the Japan Dialysis Active Vitamin D (J-DAVID) trial was performed to test the hypothesis that treatment with VDRAs reduces the risk of cardiovascular events and mortality in patients with CKD undergoing hemodialysis.

Key Points

Question Does treatment with active vitamin D reduce cardiovascular risk in patients undergoing hemodialysis?

Findings In this open-label randomized clinical trial that included 976 patients without secondary hyperparathyroidism receiving maintenance hemodialysis, the use of oral alfacalcidol compared with usual care over a median of 4 years resulted in a composite outcome of select cardiovascular events in 21.1% vs 17.9% of participants, respectively, which was not a statistically significant difference.

Meaning Active vitamin D did not reduce the risk of a composite outcome of cardiovascular events.

Methods

Ethical Considerations

This trial was conducted in accordance with the principles of the Declaration of Helsinki and the Ethical Guidelines for Clinical Studies by the Ministry of Health, Labor and Welfare, Japan (the original 2003 version, which was modified in 2004 and 2006). The protocol and revisions of this trial were approved by the ethics committee at the Osaka City University Graduate School of Medicine in Japan (approval numbers 1227, 1297, 1385, and 1525) and by the relevant ethics committees or institutional review boards at the study sites. All participants gave written informed consent before the study.

Study Design

This study was a clinical trial with a randomized, open-label, blinded end point design that examined the effect of alfacalcidol vs no VDRAs on cardiovascular events in patients receiving maintenance hemodialysis without secondary hyperparathyroidism over 48 months. The study methods have been published[16] and the trial protocol and statistical analysis plan are available in Supplement 1 and Supplement 2, respectively.

Study Population

The participants were patients aged 20 to 80 years who were receiving maintenance hemodialysis; whose serum calcium was less than or equal to 10.0 mg/dL, phosphate was less than or equal to 6.0 mg/dL, and intact PTH level was less than or equal to 180 pg/mL; and who were not taking any VDRAs at randomization. These laboratory values were derived from the 2006 version of the Clinical Practice Guideline by the Japanese Society for Dialysis Therapy (JSDT),[17] in which the recommended target intact PTH level was 60 to 180 pg/mL. Additional inclusion and exclusion criteria are listed in eTable 1 in Supplement 3. Eligible patients were recruited from July 1, 2008, to January 26, 2011, and screened at 207 dialysis centers in Japan.[16]

Randomization and Masking

Static 1:1 randomization was done using a computer-generated random sequence with a block randomization method stratified by age (<65 years vs ≥65 years), sex, years receiving dialysis (<5 years vs ≥5 years), underlying renal disease (diabetic nephropathy vs other), and history of CVD (yes vs no). Geographic region of dialysis centers was also considered in randomization, because of possible regional diversity in the lifestyle, clinical practice patterns, and risk of CVD in patients at each center. Block size was 4 for the first block and 2 thereafter because of the relatively small number of participants for each stratum. Fax was used for central randomization and communication between dialysis facilities and the data center at Osaka City University. Staff at the data center were not involved in enrollment of participants.

Intervention and Follow-up

The participants in the intervention group were assigned to receive oral alfacalcidol, starting at a dose of 0.5 μg per day, which was derived from a previous cohort study.[9] The participants in this group were asked to avoid other VDRAs, but could receive them if it was clinically necessary to adhere to the JSDT Clinical Practice Guidelines.[17,18] The participants in the control group were assigned to treatment without any VDRAs. The participants in this group were asked to avoid alfacalcidol and any other VDRA, but they were allowed to receive them if clinically necessary.

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All participants were eligible to receive any medications other than VDRAs, including phosphate binders and cinacalcet, for standard medical care as recommended by the JSDT Clinical Practice Guidelines. The 2006[7] and 2012[18] versions of these guidelines recommend that the target ranges for serum phosphate and corrected calcium concentrations should be between 3.5 and 6.0 mg/dL and between 8.4 and 10.0 mg/dL, respectively. The target range of intact PTH levels was between 60 and 180 pg/mL in the 2006 version and between 60 and 240 pg/mL in the 2012 version. The target range of intact PTH levels for participants was updated during follow-up based on contemporaneous guidelines. Serum calcium and phosphate levels of the participants were measured twice a month, and intact PTH level was measured once every 3 months at local laboratories.

In the intervention group, hypercalcemia (serum total calcium level ≥10.5 mg/dL) and hyperphosphatemia (phosphate level ≥7.0 mg/dL) were managed with dietary therapy and/or dose adjustment of calcium carbonate, sevelamer hydrochloride, or other medications. If serum calcium or phosphate levels did not improve, dose reduction or temporal cessation of the study drug was allowed. The medication was readministered after confirming the recovery of serum calcium and phosphate levels within the target ranges. The dose of alfacalcidol was allowed to be adjusted within the range of 0.25 and 7.00 μg per week. If patients' serum calcium and phosphate levels were in the target ranges, but intact PTH level exceeded the recommended limit, a switch from oral alfacalcidol to another oral or intravenous VDRA was allowed. During treatment with intravenous VDRAs, patients' intact PTH levels were measured once a month. When intact PTH was reduced to a level within the recommended range or lower, a switch back to treatment with oral alfacalcidol was considered. For the intervention group, dropout from the assigned treatment was defined as no treatment with alfacalcidol for more than 12 consecutive weeks.

For patients in the control group, if serum calcium and phosphate levels were in the recommended ranges, but intact PTH levels exceeded the recommended limit, use of oral or intravenous VDRAs was allowed. During treatment with intravenous VDRAs, intact PTH levels were measured once a month. When the intact PTH level was reduced to be within the recommended range or lower, cessation of VDRA treatment was considered. For the control group, dropout from the assigned treatment was defined as treatment with any VDRA for more than 12 consecutive weeks.

Study Outcomes and Adverse Events

The primary outcome was defined as the composite of (1) fatal and nonfatal cardiovascular events, including acute myocardial infarction, congestive heart failure requiring hospitalization, stroke, aortic dissection/rupture, amputation of ischemic limb, and sudden cardiac death; (2) coronary interventions (eg, balloon angioplasty, stenting) or bypass grafting; and (3) lower limb artery interventions (eg, balloon angioplasty, stenting) or bypass grafting. The definitions of the individual cardiovascular events are listed in eTable 2 in Supplement 3. The secondary outcome was all-cause death. These outcomes were analyzed within the fixed period of 48 months. For adverse event assessment, laboratory data and all serious adverse events (SAEs) were reported. SAEs were defined previously[16] and were categorized into cardiovascular, infection, malignancy, fall/bone fracture, accident, and other. The number of reported SAEs were summarized for each group. The event evaluation committee prospectively adjudicated the primary and secondary outcomes, as well as SAEs, with allocation groups being masked to the committee.

Sample Size Determination

To detect the difference in the proportion of the occurrence of the primary outcome between the groups with 80% power (β of 0.2) and an α of .05, we initially calculated a sample size of 1600 participants using a formula indicated in eTable 3 in Supplement 3, assuming that (1) 32% of participants in the control group would experience the primary outcome during the 4-year period,[19] (2) the risk was reduced by 20% in the intervention group, and (3) 30 participants were lost to follow-up. We took 20% risk reduction as a clinically important difference because in a population without kidney disease, a 20% risk reduction is an accepted level for the prevention of cardiovascular disease.[20] However, because of the difficulty in reaching this target sample size and possible overestimation of cardiovascular risk in the control group, we reset the target sample size to 972,[16] assuming that (1) 28% of participants in the control group would experience the primary outcome, (2) there would be a risk reduction of 30% in the intervention group, which was based on previous cohort studies of the use of oral VDRAs in patients undergoing hemodialysis,[21] and (3) 5% of participants would be lost to follow-up.

Interim Analysis

It was initially planned to perform interim analysis once, upon request of the independent data monitoring committee.[16] However, after monitoring the interim data, the committee recommended to continue the study without performing formal interim analysis because the adverse events profile, judged by the numbers of SAEs, was similar between the groups and the total number of participants who experienced the primary outcome was too small.

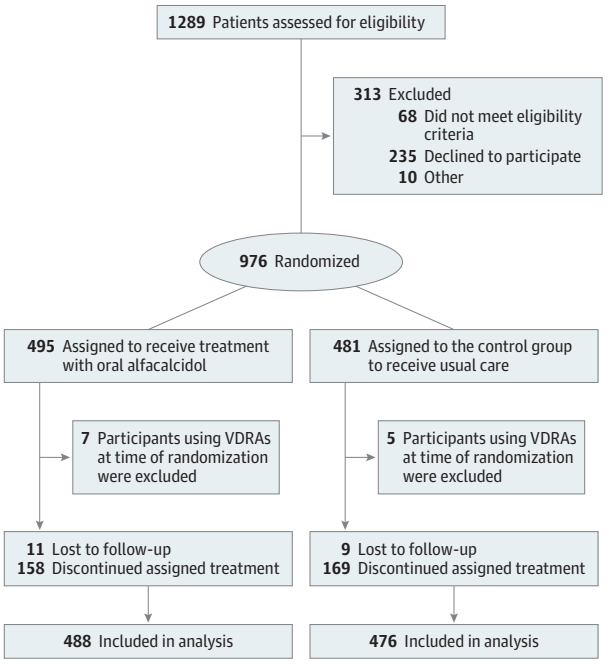
Statistical Analysis

We defined 3 populations for analysis: a full analysis set, a per-protocol analysis set, and a modified per-protocol analysis set[16] (eFigure 1 in Supplement 3). The primary analysis was performed using the time after randomization to the occurrence of the primary outcome with the full analysis set using Kaplan-Meier analysis with a log-rank test. Hazard ratio and 95% CI was calculated with an unadjusted Cox proportional hazards model for the primary composite outcome. The proportionality assumption was tested by log-minus-log plots and Schoenfeld residual tests, and we confirmed the assumption was met. Hazard ratios were also calculated for the individual components of the primary outcome. The key secondary analysis was done using time after randomization to all-cause death with the full analysis set, based on the same methods described above. As additional analyses,

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Figure 1. Flow of the Participants in a Study of the Effect of Oral Alfacalcidol in Patients Receiving Hemodialysis



The participants who were lost to follow-up were censored on the last day of follow-up. No imputation was performed. The numbers analyzed are the same for the full analysis set, the per-protocol set, and the modified per-protocol set. These analysis sets differ only in the method of censoring of participants who discontinued their assigned treatment. Such participants were censored at the time of discontinuation in the per-protocol set, whereas participants in the intervention group who switched from oral alfacalcidol to another vitamin D receptor activator (VDRA) were not censored in the modified per-protocol set.

the primary and secondary outcomes were analyzed with the per-protocol set and the modified per-protocol set (see eFigure 1 in Supplement 3 for the definitions of the analysis populations) with and without adjustment for age, sex, diabetic nephropathy, duration of hemodialysis, and history of cardiovascular disease. As a post hoc analysis, a mixed-effects model was used to further adjust for regions of the dialysis centers (sites) as a random effect. Two-sided *P* values <.05 were considered statistically significant. No imputation was done. We used SPSS version 22 (IBM Japan) and R software version 3.5.1 (R Foundation for Statistical Computing) for statistical tests.

Results

Flow of the Participants and Their Baseline Characteristics

The participant recruitment was open in July 2008, the first patient was enrolled in August 18, 2008, and the last patient was enrolled January 26, 2011. The last day of participant follow-up was April 4, 2015. Eligibility was assessed in 1289 patients, and 976 patients from 108 dialysis centers were enrolled and randomized (**Figure 1**). Of the 976 participants who were randomized, 7 patients in the intervention group

Table 1. Baseline Characteristics of Participants in a Study of the Effect of Oral Alfacalcidol in Patients Receiving Hemodialysis

Characteristic	Oral Alfacalcidol Group (n = 488)	Control Group (n = 476)
Age, median (IQR), y	65 (58-71)	65 (58-71)
Men, No. (%)	301 (61.7)	277 (58.2)
Women, No. (%)	187 (38.3)	199 (41.8)
Duration of hemodialysis, mean (IQR), y	6 (2-10)	5 (2-11)
Diabetic nephropathy, No. (%)	207 (42.4)	204 (42.9)
History of cardiovascular disease, No. (%)	127 (26.0)	117 (24.6)
Blood pressure		
Systolic, No.	486	476
Median (IQR), mm Hg	145 (130-160)	148 (134-160)
Diastolic, No.	486	475
Median (IQR), mm Hg	74 (67-82)	74 (68-83)
Pulse rate, bpm	72 (66-78)	72 (66-78)
Dialysis, median (IQR), h per wk	12 (12-12)	12 (12-12)
Single pool Kt/V[a]		
No.	477	465
Median (IQR)	1.41 (1.23-1.61)	1.44 (1.20-1.64)
Dialysate calcium concentration, No. (%), mEq/L		
2.5	130 (26.6)	121 (25.4)
3.0	333 (68.2)	331 (69.5)
Other	25 (5.1)	24 (5.0)
Height, cm		
No.	478	468
Median (IQR)	160 (154-167)	161 (153-167)
Body weight, kg		
No.	487	
Median (IQR)	54.5 (48.2-61.5)	53.9 (47.5-61.9)
BMI		
No.	477	468
Median (IQR)	21.1 (19.0-23.4)	21.1 (19.1-23.3)
Laboratory parameters, median (IQR)		
C-reactive protein, mg/dL		
No.	427	414
Median (IQR)	0.10 (0.05-0.30)	0.10 (0.05-0.26)
Serum albumin, g/dL		
No.	487	476
Median (IQR)	3.8 (3.5-3.9)	3.8 (3.5-4.0)
Serum calcium, mg/dL	8.9 (8.5-9.2)	8.8 (8.5-9.2)
Corrected calcium, mg/dL	9.1 (8.8-9.5)	9.1 (8.7-9.5)
Phosphate, mg/dL	4.6 (3.9-5.2)	4.7 (4.0-5.4)
Intact parathyroid hormone, pg/mL	85.1 (45.2-130.0)	86.1 (47.0-127.4)
Hemoglobin, g/dL	10.6 (10.0-11.3)	10.7 (10.1-11.3)
Creatinine, mg/dL	10.6 (8.9-12.4)	10.4 (8.7-12.1)
Blood urea nitrogen, mg/dL	64 (55-74)	64 (55-76)
Sodium, mEq/L	140 (138-142)	140 (138-142)
Potassium, mEq/L	4.8 (4.3-5.3)	4.8 (4.2-5.3)

(continued)

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Table 1. Baseline Characteristics of Participants in a Study of the Effect of Oral Alfacalcidol in Patients Receiving Hemodialysis (continued)

Characteristic	Oral Alfacalcidol Group (n = 488)	Control Group (n = 476)
Chloride, mEq/L		
No.	486	471
Median (IQR)	103 (101-106)	103 (101-106)
Total cholesterol, mg/dL		
No.	480	464
Median (IQR)	152 (133-176)	150 (130-170)
Triglycerides, mg/dL		
No.	449	439
Median (IQR)	97 (68-136)	96 (70-134)
HDLC, mg/dL		
No.	447	434
Median (IQR)	46 (38-56)	46 (37-55)
AST, IU/L		
No.	487	474
Median (IQR)	14 (10-18)	14 (10-17)
ALT, IU/L		
No.	487	474
Median (IQR)	11 (8-14)	10 (8-14)
Hemoglobin A1c,[b] %		
No.	186	191
Median (IQR)	6.1 (5.6-6.9)	6.1 (5.7-7.0)
Glycated albumin,[b] %		
No.	144	159
Median (IQR)	20.9 (18.3-24.7)	21.2 (18.9-25.3)
Use of medication for MBD in CKD, No. (%)		
Calcium carbonate	412 (84.4)	392 (82.4)
Sevelamer hydrochloride	153 (31.4)	155 (32.6)
Lanthanum carbonate	69 (14.1)	49 (10.3)
Cinacalcet hydrochloride	27 (5.5)	29 (6.1)
Active vitamin D sterol	0 (0.0)	0 (0.0)
Use of medication for renal anemia, No. (%)		
Epoetin injection	209 (42.8)	212 (44.5)
Long-acting ESA	171 (35.0)	162 (34.0)
Intravenous iron	112 (23.0)	118 (24.8)
Use of medication for dyslipidemia, No. (%)		
Statin	77 (15.8)	81 (17.0)
Use of medication for hypertension, No. (%)		
Calcium channel blocker	246 (50.4)	262 (55.0)
ACEI	41 (8.4)	41 (8.6)
ARB	237 (48.6)	245 (51.5)
β-blocker	109 (22.3)	123 (25.8)
Loop diuretics	141 (28.9)	127 (26.7)
Use of medication for diabetes mellitus, No. (%)		
Sulfonyl urea	10 (2.1)	4 (0.8)
Rapid-acting insulin secretagogue	17 (3.5)	13 (2.7)
α-glucosidase inhibitor	49 (10.0)	50 (10.5)
Insulin injection	77 (15.8)	88 (18.5)
Use of antithrombotic agents, No. (%)		
Warfarin	30 (6.2)	31 (6.5)
Aspirin	149 (30.5)	128 (26.9)
Ticlopidine	31 (6.4)	21 (4.4)
Cilostazol	48 (9.8)	35 (7.4)

(continued)

Table 1. Baseline Characteristics of Participants in a Study of the Effect of Oral Alfacalcidol in Patients Receiving Hemodialysis (continued)

Abbreviations: ACEI, angiotensin-converting enzyme inhibitor; ARB, angiotensin II receptor blocker; AST, aspartate aminotransferase; ALT, alanine aminotransferase; BMI, body mass index (calculated as weight in kilograms divided by height in meters squared); CKD, chronic kidney disease; ESA, erythrocyte stimulating agent; HDLC, high-density lipoprotein cholesterol; IQR, interquartile range; MBD, mineral and bone disorder.

[a] Single pool Kt/V was an indicator of dialysis adequacy calculated by dialyzer clearance of urea (K), dialysis time (t), and volume of distribution of urea (V) using a single pool model.

[b] Reported only for patients with diabetic nephropathy.

and 5 in the control group were excluded just after randomization because they were using VDRAs at the time of randomization. No follow-up data were available for these participants. Thus, the full analysis set included 488 participants in the intervention group and 476 in the control group. **Table 1** shows the baseline characteristics of the participants. There was complete loss to follow-up of 11 and 9 participants in the intervention and control group, respectively, and these participants were censored on the last day of follow-up. The presence or absence of the primary outcome was known in 944 of 964 participants (97.9%) in the full analysis set. Assigned treatment was discontinued in 158 and 169 participants in the intervention group and the control group, respectively, who were censored, as defined, in the per-protocol set and the modified per-protocol set.

Changes in Laboratory Data, Medications, and Adherence

In the control group, there was no apparent change in serum phosphate or corrected calcium levels during follow-up. Intact PTH levels slightly increased over time, but remained within the target range (eFigure 2 in Supplement 3). In contrast, the intervention group showed an initial increase in serum corrected calcium and an initial decrease in intact PTH levels, but these levels returned to the initial levels over time.

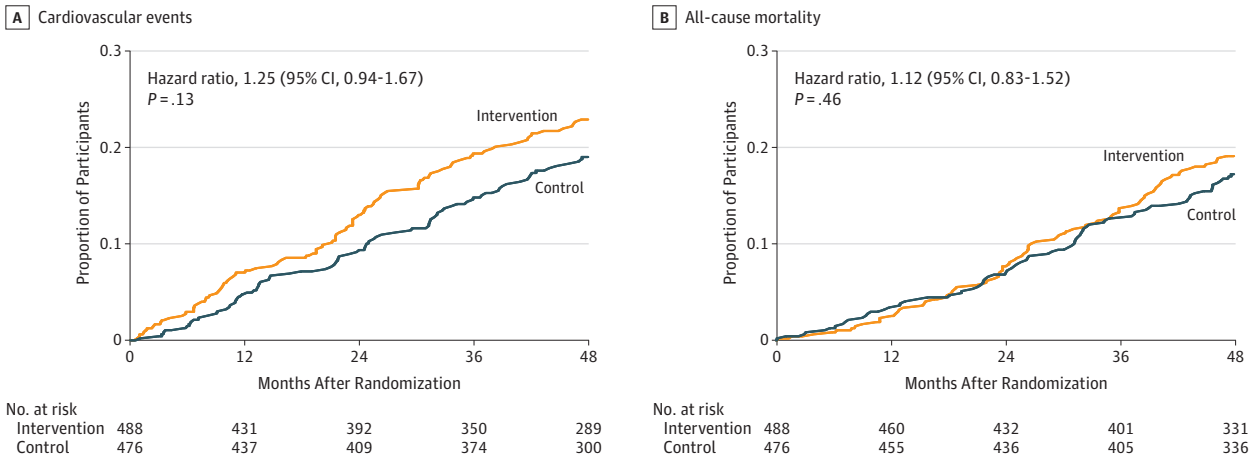
There were changes in medications in both groups (eFigure 2 in Supplement 3). Compared with the control group, the intervention group had lower proportions of calcium carbonate users and cinacalcet users, and higher proportions of sevelamer users and lanthanum carbonate users. The proportion of intravenous VDRA users increased over time in both groups, although the proportion was higher in the control group.

The proportion of participants who received the assigned treatment declined over time similarly in the groups (eFigure 3 in Supplement 3). Treatment with alfacalcidol was stopped for 158 of 488 patients (32.4%) in the intervention group, whereas treatment with any VDRA was started for 169 of 476 patients (35.5%) in the control group.

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Figure 2. Primary and Secondary Outcomes



A, For the primary outcome, the median (interquartile range [IQR]) observation time was 4.0 (2.6-4.0) years for the intervention group and 4.0 (3.5-4.0) years for the control group. B, For the secondary outcome, the median (IQR) observation time was 4.0 (4.0-4.0) years for the intervention group and 4.00 (4.0-4.0) years for the control group. The primary outcome (A) included fatal and nonfatal cardiovascular events, coronary revascularization, and leg artery revascularization. The secondary outcome (B) was all-cause mortality.

Table 2. Primary and Secondary Outcomes and Serious Adverse Events in Participants Undergoing Hemodialysis Who Received Alfacalcidol or Standard Care

	Intervention Group (n = 488)	Control Group (n = 476)	Absolute Difference, % (95% CI)	Hazard Ratio (95% CI)
Composite first cardiovascular events, No. (%)	103 (21.1)	85 (17.9)	3.25 (−1.75 to 8.24)	1.25 (0.94-1.67)
Cardiovascular disease, No. (%)	67 (13.7)	56 (11.8)	1.96 (−2.24 to 6.17)	1.26 (0.88-1.79)
Acute myocardial infarction, No.	10	11		
Congestive heart failure, No.	15	14		
Stroke, No.	28	14		
Aortic dissection/Rupture, No.	3	2		
Amputation of ischemic limb, No.	4	4		
Cardiac sudden death, No.	7	11		
Coronary intervention/bypass, No. (%)	21 (4.3)	18 (3.8)	0.52 (−1.96 to 3.01)	1.20 (0.64-2.25)
Percutaneous intervention, No.	20	11		
Bypass grafting, No.	1	7		
Leg artery intervention/bypass, No. (%)	15 (3.1)	11 (2.3)	0.76 (−1.28 to 2.80)	1.40 (0.64-3.05)
Percutaneous intervention, No.	14	11		
Bypass grafting, No.	1[a]	0		
All-cause mortality (secondary outcome), No. (%)	89 (18.2)	80 (16.8)	−0.25 (−5.14 to 4.34)	1.12 (0.83-1.52)
Serious adverse events, No. (%)	371 (76.0)	377 (79.2)		
Cardiovascular, No.	199	191		
Infection, No.	64	63		
Malignancy, No.	22	21		
Falls/Bone fracture, No.	9	12		
Other,[b] No.	77	90		

[a] One patient who received both leg artery bypass grafting and percutaneous intervention at the same time was classified as being treated with bypass grafting in this table.

[b] Including respiratory, hepatobiliary, gastrointestinal, urogenital, dermal, ophthalmologic, and orthopedic adverse events and accidents.

Primary and Secondary Outcomes

During follow-up (median, 4.0 years; mean, 3.6 years), the primary outcome occurred in 103 patients (21.1%) in the intervention group and 85 patients (17.9%) in the control group (absolute difference, 3.25% [95% CI, −1.75% to 8.24%]; hazard ratio, 1.25 [95% CI, 0.94-1.67]; P = .13). There was no significant difference between the intervention and control group in the secondary outcome of all-cause death (18.2% vs 16.8%, respectively; hazard ratio, 1.12 [95% CI, 0.83-1.52]; P = .46). **Figure 2** shows the Kaplan-Meier curves for the primary and secondary outcomes from the intention-to-treat analysis with the full analysis set. **Table 2** summarizes the composite and individual cardiovascular events that occurred.

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Table 3. Primary and Secondary Outcomes in Participants Undergoing Hemodialysis Who Received Alfacalcidol or Standard Care[a]

Outcome and Analysis Method	No. (%)		Unadjusted		Adjusted 1[b]		Adjusted 2[c]	
	Intervention Group (n = 488)	Control Group (n = 476)	Hazard Ratio (95% CI)	*P* Value	Hazard Ratio (95% CI)	*P* Value	Hazard Ratio (95% CI)	*P* Value
Cardiovascular Events								
Full analysis set	103 (21.1)	85 (17.9)	1.25 (0.94-1.67)	.13	1.26 (0.95-1.68)	.11	1.29 (0.96-1.72)	.09
Per-protocol set	87 (17.8)	68 (14.3)	1.32 (0.96-1.81)	.09	1.32 (0.96-1.82)	.09	1.35 (0.98-1.86)	.07
Modified per-protocol set	89 (18.2)	68 (14.3)	1.33 (0.97-1.82)	.08	1.34 (0.97-1.83)	.07	1.36 (0.99-1.87)	.06
All-Cause Mortality								
Full analysis set	89 (18.2)	80 (16.8)	1.12 (0.83-1.52)	.46	1.12 (0.83-1.52)	.46	1.13 (0.83-1.53)	.43
Per-protocol set	66 (13.5)	67 (14.1)	0.98 (0.70-1.38)	.91	0.97 (0.69-1.36)	.86	0.97 (0.69-1.36)	.85
Modified per-protocol set	67 (13.7)	67 (14.1)	0.98 (0.70-1.37)	.89	0.97 (0.69-1.36)	.86	0.97 (0.69-1.36)	.85

[a] The primary intention-to-treat analysis was done for the primary outcome with the full analysis set. Additional analysis was performed with the per-protocol set and the modified per-protocol set. The per-protocol set consists of all eligible participants who were randomized, but participants were censored at the time of discontinuation of the assigned treatment. The modified per-protocol set differs from the per-protocol set in the respect that participants in the intervention group who switched from alfacalcidol to another oral/intravenous vitamin D receptor activators were not censored.

[b] Adjusted for age, sex, dialysis duration, underlying diabetic nephropathy, and history of cardiovascular disease.

[c] Adjusted with a mixed-effects model (post hoc) with regions (sites) as a random effect in addition to age, sex, dialysis duration, underlying diabetic nephropathy, and history of cardiovascular disease as fixed effects.

Adverse Events

As shown in Table 2, the number of SAEs was similar between the groups. The number of cardiovascular SAEs was higher than the number of occurrences of the primary outcome because some participants had more than 2 cardiovascular SAEs. The predefined laboratory abnormalities are summarized in eTable 4 in Supplement 3, and show that corrected serum calcium greater than 10.0 mg/dL and phosphate greater than 6.0 mg/dL were more common in the intervention group than the control group, and intact PTH greater than 240 pg/mL was less common, particularly in the first year of follow-up.

Additional Analyses

The adjusted hazard ratios for the primary outcome were 1.32 (95% CI, 0.96-1.82; *P* = .09) and 1.34 (95% CI, 0.97-1.83; *P* = .07) when the per-protocol set and the modified per-protocol set, respectively, were analyzed. The adjusted results of the preplanned and post hoc analyses are shown in **Table 3.**

Discussion

In patients without secondary hyperparathyroidism receiving maintenance hemodialysis, oral alfacalcidol compared with usual care did not reduce the risk of a composite measure of select cardiovascular events.

The PRIMO[22] and OPERA[23] studies showed no significant effect of oral paricalcitol on left ventricular mass index in patients with CKD. Three 2017 studies that examined the effects of calcitriol and nutritional vitamin D sterols on vascular functions, measured by arterial stiffness[24] or flow-mediated dilatation,[25,26] reported inconsistent results. The current trial failed to show the cardiovascular benefit of VDRAs in patients with CKD receiving hemodialysis.

Despite the lack of evidence by randomized trials, many nephrologists consider the administration of VDRAs to be mandatory for patients undergoing hemodialysis, based on a number of observational studies that showed associations between VDRA use and lower risk of all-cause mortality[8] and cardiovascular outcomes[9,10,21] in participants receiving hemodialysis. A trial with VDRAs was deemed not to be feasible by many nephrologists because half of the included patients would not be treated with VDRAs during the trial. This issue was addressed in the current trial by selecting patients with intact PTH levels below the maximum limit of the target range recommended by the JSDT, for whom VDRAs were not necessarily required.

The hazard ratio of 1.25 for the primary outcome derived from intention-to-treat analysis was not statistically significant. However, the hazard ratio was higher when analyzed using the per-protocol and modified per-protocol sets with preplanned or post hoc adjustment. The results of the additional analyses should be interpreted cautiously, but they may be a signal of the potential harm rather than benefit of VDRAs in patients without elevated serum intact PTH levels. VDRAs increase serum FGF23, which induces left ventricular hypertrophy[27] and is associated with congestive heart failure[28] and mortality.[29] In the current trial, the occurrence of congestive heart failure and cardiac sudden death was similar between the groups. The occurrence of coronary events was also similar. The occurrence of stroke was twice as high in the intervention group than the control group (28 vs 14). Although this observation may be coincidental, the results raise the possibility that MBD in CKD and/or methods used to manage it could influence the cerebrovascular system.

The results of this trial were different than the results of previous observational studies. There are possible explanations for this discrepancy. First, parathyroid function or bone turnover may modify the cardiovascular effects of VDRAs. Typically, VDRAs are prescribed to patients with hyperparathyroidism and high bone turnover, whereas alfacalcidol was given to patients with serum intact PTH levels less than or equal to 180 pg/mL in the current trial. VDRAs may only benefit patients with secondary hyperparathyroidism by suppressing phosphate/calcium mobilization from bone. Second, the cardiovascular effects of VDRAs may vary depending on the degree of calcium load. In this trial, more than 68% of the par-

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ticipants received dialysis against 3.0 mEq/L calcium, and more than 80% of the participants were treated with calcium carbonate at baseline. Third, observational studies could be confounded by some important but unmeasured variables for which statistical adjustment cannot be made. Gene polymorphisms affect bone turnover and response to calcium and vitamin D supplementation.[30] Patients not using VDRAs in cohort studies may be the patients who could not continue VDRAs in these studies because of hypercalcemic response.

Limitations

This study has several limitations. First, the statistical analysis may have been underpowered for the primary and secondary outcomes. Based on the actual number of patients that experienced the primary outcome in the control group, the power to detect a 20% risk reduction was calculated to be approximately 24%. The null findings are not definitive. Second, because of the unblinded design, there are some potential biases, such as unequal cointerventions and assessment of outcomes in favor of the hypothesis. Although adjudication of outcomes was done by a blinded committee, reporting bias by investigators cannot be excluded. However, such biases are not likely to affect the results, because the results were against the study hypothesis. Third, there was considerable contamination by dropout and readministration of VDRA use, which could reduce statistical power and affect the results. However, the additional analysis with the per-protocol set and modified per-protocol set did not support the cardiovascular benefit of treatment with VDRAs. Fourth, information on serum 25-hydroxyvitamin D level or consumption of native vitamin D supplements are not available. Fifth, the results cannot be generalized to patients with secondary hyperparathyroidism or to non-Japanese populations, particularly not to US patients undergoing hemodialysis, who have much higher levels of intact PTH than the participants of this trial. Sixth, no information was recorded on why the participants discontinued their treatment. Although the reasons are likely to be laboratory abnormalities, such as high serum calcium in the intervention group and high intact PTH levels in the control group, no data are available in case report forms.

Conclusions

Among patients without secondary hyperparathyroidism undergoing maintenance hemodialysis, oral alfacalcidol compared with usual care did not reduce the risk of a composite measure of select cardiovascular events. These findings do not support the use of vitamin D receptor activators for patients such as these.

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Additional Information: This trial was designed and overseen by the principal investigator and the steering committee, which nominated the executive committee to organize the J-DAVID Research Group composed of 207 dialysis facilities to recruit participants. The data center played roles in data compilation, data cleaning, and the central monitoring of the trial. The event evaluation committee adjudicated the primary and secondary outcomes and SAEs. The independent data monitoring committee reviewed outcome and adverse events data. A sampling source data verification and audit was performed as described previously.[16]

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